

A normal distribution

A normal (Gaussian) distribution is continuous, symmetric, and bell shaped, with mean μ and standard deviation σ .

What does moving μ change about the curve?

Normal notation

In this textbook, the second entry in $N(\mu, \sigma)$ is the standard deviation.

$$X \sim N(\mu, \sigma)$$

For $X \sim N(40, 6)$, what are the mean and standard deviation?

The standard normal

The standard normal distribution has mean zero and standard deviation one.

$$Z \sim N(0, 1)$$

Where is the center of this distribution?

A standardized value

A z-score measures distance from the mean in standard deviations.

$$z = \frac{x - \mu}{\sigma}$$

What does a z-score of -1.5 tell us?

A positive z-score

Model: component lengths are normal with mean 40 mm and standard deviation 6 mm.

What is the z-score for a length of 49 mm?

A negative z-score

Model: component lengths are normal with mean 40 mm and standard deviation 6 mm.

What is the z-score for a length of 28 mm?

At the mean

Model: X is normal with mean 40 and standard deviation 6.

What z-score belongs to $X=40$?

Returning to original units

A value can be recovered from its z-score, mean, and standard deviation.

$$X = \mu + z\sigma$$

For $\mu=40$ and $\sigma=6$, what value has $z=-0.5$?

Comparing different scales

Model A has mean 40 and SD 6. Model B has mean 100 and SD 12.

Which is farther above its mean in standard deviations: A=49 or B=112?

Scores are not probabilities

A z-score describes standardized location.

Why can $z=2.5$ be valid while a probability of 2.5 cannot?

Standardizing an event

Model: $X \sim N(40, 6)$.

What standard-normal event corresponds to $X < 46$?

Symmetry

The standard normal density is symmetric about zero.

How does $P(Z < -1.4)$ compare with $P(Z > 1.4)$?

The center divides the area

Model: Z is standard normal.

What is $P(Z < 0)$?

One standard deviation

For a normal distribution, about 68% of values lie within one standard deviation of the mean.

For $\mu=40$ and $\sigma=6$, what interval contains that central region?

Empirical Rule

The Empirical Rule, or 68–95–99.7 rule, gives approximate central areas within one, two, and three normal standard deviations.

For $\mu = 40$ and $\sigma = 6$, what endpoints enclose the middle 99.7%?

Two standard deviations

For a normal distribution, about 95% of values lie within two standard deviations of the mean.

Approximately what proportion lies above $\mu + 2\sigma$?

Three standard deviations

For a normal distribution, about 99.7% of values lie within three standard deviations of the mean.

Approximately what proportion lies outside that interval?

Areas between standard deviations

Use the 68–95–99.7 rule for a normal distribution.

What percentage lies between one and two standard deviations above the mean?

Using a cumulative table

A standard-normal table may report area to the left of z .

If its entry for $z=1.2$ is 0.8849, what is $P(Z>1.2)$?

A negative cumulative cutoff

Given: $P(Z < 1.2) = 0.8849$ for a standard normal Z .

What is $P(Z < -1.2)$?

Normality matters

Synthetic data have a long right tail and several unusually large values.

Would calculating z-scores alone justify standard-normal probability calculations?

Location versus shape

Standardization subtracts a mean and divides by a positive standard deviation.

Would standardizing a strongly skewed variable make its distribution normal?